Data Mining – Lecture 4

1-Merging data from multiple data stores help reduce and avoid ___ in the resulting data set.

a- Redundancies
b- Inconsistencies
c- Both
d- None

Answer : c

2- _____ is a challenge of data integration

a- Semantic heterogeneity
b- Structure of data
c- Redundancy
d- All above

Answer : d.

3- ______ is related to entity identification problem.

a- Semantic heterogeneity
b- Structure of data
c- Redundancy
d- All above

Answer : a.

4- _____ can help to avoid errors in schema integration and data transformation.

a- Metadata
b- Gigadata
c- Teradata
d- None

Answer : a.

5- Redundancy can be detected by correlation analysis which measures how strongly one attribute implies the other , based on the available data.

a- True
b- False

Answer : a.

6- ______ is used for nominal data.

a- Correlation coefficient
b- Covariance
c- Chi-square
d- None

Answer : c . example (gender)

7- _____ is used for numeric attributes .

a- Correlation coefficient
b- Covariance
c- Chi-square
d- A and B

Answer : d. example

		gender		
		male	female	Total
Preferred reading	Fiction	250	200	450
	Non-fiction	50	1000	1050
	Total	300	1200	1500

"The upcoming questions are related to this example "

8- What is the expected frequency for e_{11} (Male-fiction)

a- 90
b- 360
c- 210
d- 1000

Answer : a.

9- What is the expected frequency for e_{12} (female-fiction)

a- 90
b- 360
c- 210
d- 1000

Answer : b.

10- Find Chi-square for the data = ____ .

a- 505.93
b- 507.93
c- 500.93
d- None

Answer : b.

Data Mining – Lecture 4

$-1 \leq r_{A,B} \leq +1$

Related to Correlation coefficient for numeric attributes .

11- if the Correlation coefficient of A and B is equal to 0.5 then it's a ____ correlated.

a- Positively
b- Negatively
c- None
d- Both

Answer : a.

12- if the Correlation coefficient of A and B is equal to Zero then it's a ____ correlated.

a- Positively
b- Negatively
c- Not (independent)
d- None

Answer : c.

13- if the Correlation coefficient of A and B is equal to 1.6 then it's a ___ correlated.

a- Positively
b- Negatively
c- Not (independent)
d- Error

Answer : d.

14- Correlation does not imply causality

a- True
b- False

Answer : a.

15- If the Covariance of A and b is equal to 1.6 then it's a ____ correlated.

a- Positively
b- Negatively
c- Not (independent)
d- Error

Answer : a.

Note (Covariance is either Zero or negative or positive)
Note (Correlation coefficient is either from -1 to 0 to 1)

Data Mining – Lecture 4

16- ____ reduction , reduces the number of attributes.
- a- Numerosity
- b- Dimensionality
- c- Both
- d- None

Answer : b.

17- Numerosity reduction replaces original data volume by smaller data representation .
- a- True
- b- False

Answer : a.

18____ is an example of Nonparametric numerosity reduction.
- a- Histograms
- b- Clustering
- c- Sampling
- d- All above

Answer : d.

19- Regression is a parametric numerosity reduction.
- a- True
- b- False

Answer : a.

20- What is the primary advantage of using heuristic (greedy) search in attribute subset selection?
- a- It guarantees finding the optimal solution.
- b- It is computationally efficient compared to exhaustive search.
- c- It requires less memory resources.
- d- It is less prone to overfitting.

Answer : b. note (exhaustive search can be prohibitively expensive as well “

21- Which attribute subset selection method involves starting with the full set of attributes and iteratively removing the worst attribute?
- a- Stepwise forward selection
- b- Decision tree induction
- c- Stepwise backward elimination
- d- Attribute construction

Answer : c.

22- What approach combines both forward selection and backward elimination in attribute subset selection?

a- Decision tree induction
b- Combination of forward selection and backward elimination
c- Attribute construction
d- Heuristic (Greedy) search

Answer : b.

23- Which method aims to find a minimum set of attributes while preserving the original probability distribution of data?

a- Decision tree induction
b- Stepwise backward elimination
c- Attribute construction
d- Stepwise forward selection

Answer : d.

24- What is a characteristic of attribute subset selection through stepwise backward elimination?

a- It starts with an empty set of attributes.
b- It involves adding the best attribute at each iteration.
c- It removes the worst attribute at each iteration.
d- It constructs new attributes based on existing ones.

Answer : c.

25- In regression analysis, what does the regression line equation y = wx + b represent?

a- The line of best fit for the data
b- The equation for a parabolic curve
c- The sum of squared errors
d- The correlation coefficient between variables

Answer : a.

26- What is a characteristic of equal-width histograms?

a- The frequency of each bucket is constant.
b- Buckets represent continuous ranges for the attribute.
c- Each bucket contains roughly the same number of data samples.
d- The width of each bucket range is uniform.

Answer : d.

Data Mining – Lecture 4

27- Equal-frequency histograms ensure roughly constant frequency per bucket.
- a- True
- b- False

Answer : a.

28- What is the primary difference between simple random sample with replacement (SRSWR) and simple random sample without replacement (SRSWOR)?
- a- SRSWOR ensures that each tuple has an equal probability of being sampled.
- b- SRSWR allows each tuple to be drawn multiple times.
- c- SRSWOR records each sampled tuple and removes it from the population.
- d- SRSWR divides tuples into strata before sampling.

Answer : b.

29- In data transformation, what does normalization involve?
- a- Dividing data into clusters or groups.
- b- Replacing raw values of numeric attributes with interval or conceptual labels.
- c- Converting data into a more structured form.
- d- Combining multiple attributes into a single attribute.

Answer : b.

30- Which transformation strategy involves replacing the raw values of a numeric attribute with interval labels or conceptual labels?
- a- Smoothing
- b- Aggregation
- c- Normalization
- d- Attribute construction

Answer : c.

31- What is the purpose of normalization in data transformation?
- a- To combine multiple attributes into a single attribute.
- b- To convert categorical data into numerical data.
- c- To generalize lower-level concepts into higher-level concepts.
- d- To scale numeric attributes to a standard range or distribution.

Answer : d.

32- What does the z-score normalization method normalize attribute values based on?
- a- The maximum and minimum values of the attribute.
- b- The mean and standard deviation of the attribute.
- c- The median and interquartile range of the attribute.
- d- The mode and variance of the attribute.

Answer : b.

$$\acute{v}_i = \frac{v_i - \bar{A}}{\sigma_A}$$

Example: income *mean and SD* are $54000 and $16000, a z-score for a value of $73600 for income is

$$\frac{73600-54000}{16000} = 1.225$$

33- For an income ranging from $12000 to $98000, mapped to the range [1, 3], what is the min-max normalization for an income of $73600?

a- 2.431
b- 1.569
c- 1.568
d- 1.431

$$\acute{v}_i = \frac{v_i - min_A}{max_A - min_A}(new\_max_A - new\_min_A) + new\_min_A$$

Answer : a , Note (focus on this questions and specifcaly on some points , such as the new min value which is 1)

34- What is the primary purpose of concept hierarchy in data transformation and discretization?

a- To randomly organize attribute values.
b- To reduce the complexity of data by organizing concepts hierarchically.
c- To ensure all attributes have equal weight.
d- To combine multiple attributes into a single attribute.

Answer : b.

35- How are concept hierarchies formed for numeric data?

a- By specifying a partial ordering of attributes explicitly by users or experts.
b- By automatically generating concept hierarchies based on distinct values per attribute.
c- By randomly selecting attribute values.
d- By transforming numeric data into nominal data.

Answer : b.

36- What is one advantage of concept hierarchies in data analysis?

a- They introduce more complexity into the data.
b- They facilitate drilling and rolling to view data in multiple granularity.
c- They make data analysis slower and more cumbersome.
d- They limit the ability to explore data at different levels of detail.

Answer : b.

37- Concept hierarchy formation reclusively reduces' data by collecting and replacing low level concepts by higher level concepts .

a- True
b- False

Answer : a.

Data Mining – Lecture 4

38- In data transformation or discretization, what technique involves organizing concepts hierarchically to reduce data complexity?

a- Attribute subset selection
b- Concept hierarchy
c- Wavelet transforms
d- Histogram analysis

Answer : b.

39- Among the techniques listed , which one does not involve the application of regression?

a- Reduction
b- Transformation
c- Cleaning
d- Integration

Answer : d.

" شكرا لكم ولا تنسونا بالدعاء"